9	**(a)**		'light' nuclei combine to form 'heavier' nuclei	B1	[1]
	(b)	**(i)**	*either* energy = $c^2\Delta m$		
			or energy = $(3.00 \times 10^8)^2 \times 1.66 \times 10^{-27}$	C1	
			energy $= 1.494 \times 10^{-10}$ J	C1	
			$= (1.494 \times 10^{-10}) / (1.60 \times 10^{-13})$		
			= 934 MeV (*3 s.f.*)	A1	[3]
		(ii)	Δm = (2.01356 + 3.01551) – (4.00151 + 1.00867)		
			= 5.02907 – 5.01018		
			= 0.01889 u	C1	
			energy = 0.01889×934		
			= 17.6 MeV (*allow 2 s.f.*)	A1	[2]
		(iii)	high temperature means high speeds / <u>kinetic</u> energy of nuclei	B1	
			D and T nuclei collide despite repelling one another	B1	[2]

Section B